Rastera

Rastera is a small language with symbolic control flow: `@` defines a function, `?/:` is if/else, `*` is a while loop or (`* i :: a..b`) a range loop, and `^` returns. This repository implements it twice, as the two phases of the course project:

Phase	Program	What it does
1	`rasteri`	Flex + Bison interpreter. Parses a `.rst` file, checks it, and runs `@main` directly. No IR, no executable is produced.
2	`rasterc`	Flex + Bison ahead-of-time compiler. Builds an AST, runs semantic analysis, emits LLVM 18 IR and links a native Linux executable.

Rastera's own feature is built-in procedural graphics: a program draws into an in-memory canvas with `canvas`, `rgb`, `pixel`, `line` and `rect`, and `export()` writes it as a PPM image (section 5.5 and the graphics paragraph of section 6).

Both programs accept the same language and are checked by the same test cases (`tests/pass`, `tests/fail`). The normative language rules are in `docs/LANGUAGE_SPEC.md`; the original project proposal is `docs/proposal.pdf`.

```
@main() [
    print(6 * 7);
    ^ 0;
]
```

1. Requirements (Linux)

Primary path: Docker. The whole toolchain lives in one Docker image (`Dockerfile`, Ubuntu 24.04), so nothing else has to be installed. `./dev.sh` builds that image on first use and runs every command inside it. Only `docker build` and `docker run` are used, so the Docker Compose plugin is not needed.

On Ubuntu or Debian, install Docker Engine from the distribution package, and git to clone:

```
sudo apt-get update
sudo apt-get install -y docker.io git
sudo usermod -aG docker "$USER"     # then log out and back in so `docker` works without sudo
docker run --rm hello-world         # optional check that Docker works
```

(The development machine used for this README runs Ubuntu 24.04 with the `docker.io` package, Docker 29.1.3, and a user that is a member of the `docker` group.) For other distributions follow the Docker Engine installation guide for your distribution on docs.docker.com. Under Windows use a WSL2 Ubuntu terminal with Docker available inside it; the commands below are for Linux.

Optional native path (no Docker). On Ubuntu 24.04 install the same packages the `Dockerfile` installs:

```
sudo apt-get install -y build-essential cmake ninja-build git flex bison \
    llvm-18-dev clang-18 libzstd-dev zlib1g-dev libedit-dev libcurl4-openssl-dev
```

The build requires: CMake >= 3.20 (24.04 ships 3.28.3), Ninja, a C and C++17 compiler with `cc` available for linking, Flex >= 2.6 (2.6.4), Bison >= 3.8 (3.8.2) and the LLVM 18 development package (`llvm-18-dev`). Native configure and build are the same two commands `dev.sh` runs:

```
cmake -S . -B build -G Ninja
cmake --build build
```

These commands were verified inside the Docker image on a fresh `git clone`. The wrapper `./dev.sh` is the supported route; if CMake cannot find LLVM natively, add `-DLLVM_DIR=/usr/lib/llvm-18/lib/cmake/llvm` to the configure command.

2. Get, build and test

```
git clone https://github.com/MHassanAbaid24/Rastera.git
cd Rastera
./dev.sh build        # builds the image on first use, then build/rasteri and build/rasterc
./dev.sh test         # build, then run BOTH test suites
```

`./dev.sh test` runs `tests/run_tests.sh build/rasterc` (every program compiled by `rasterc` and executed) and `tests/run_tests.sh --interp build/rasteri` (every program run through `rasteri`). Expected final output (one line per suite):

```
161 passed, 0 failed
161 passed, 0 failed
```

A failing case prints `FAIL <test>: <reason>` and the suite exits non-zero. `./dev.sh` with no arguments opens a shell in the container, and `./dev.sh <command>` runs any command there. All paths in this README are relative to the repository root, which is `/work` in the container. Build outputs go to `build/`, which git ignores.

3. Phase 1: the interpreter `rasteri`

Build: `./dev.sh build` produces `build/rasteri` (CMake target `rasteri`, sources `src/interp/interp.y` and `src/interp/lexer.l`; its Flex/Bison symbols use the prefix `ip` so it can live beside the compiler's parser).

Run:

```
./dev.sh build/rasteri examples/recursion_arrays.rst
```

- The program's `print(x)` calls write to stdout, one value and a newline each.
- Static errors (syntax errors, undefined variables, ...) are reported on stderr as `FILE:LINE: error: message` and nothing runs: exit status 1.
- A runtime error stops the program with `rastera: runtime error: MESSAGE` on stderr, exit status 2. Output printed before the error is kept.
- Otherwise the exit status is the value returned by `@main` (its low 8 bits, as for any Linux process).
- `rasteri` never writes an executable and never builds IR. Wrong usage prints `usage: rasteri <file.rst>` and exits 1; a missing file prints `FILE: No such file or directory`, exit 1.

4. Phase 2: the compiler `rasterc`

Build: `./dev.sh build` produces `build/rasterc` (and the runtime library `build/librastera_rt.a` that every program links with).

```
./dev.sh build/rasterc examples/recursion_arrays.rst -o build/recursion_arrays   # compile
./dev.sh build/recursion_arrays                                                   # run
```

Without `-o` the executable is named `a.out`. Stages can be inspected instead of building:

```
./dev.sh build/rasterc FILE.rst --tokens   # lexer output only
./dev.sh build/rasterc FILE.rst --ast      # parse only, print the AST
./dev.sh build/rasterc FILE.rst --ir       # stop after code generation, print LLVM IR
```

- Exit status of `rasterc` itself: 0 on success; 1 for a missing file, bad usage or any static error (`FILE:LINE: error: message` on stderr, no executable written). Usage text: `usage: rasterc <file.rst> [-o <output>] [--tokens | --ast | --ir]`.
- `--ast` stops after parsing, so it also prints programs that semantic analysis would reject (see `examples/static_error.rst`).
- The produced executable is a normal Linux program: `print` writes to stdout, the exit status is `@main`'s result, and a runtime error prints `rastera: runtime error: MESSAGE` on stderr and exits 2.

Real inspection output for `tests/pass/return_literal.rst` (`@main() [ ^ 42; ]`):

```
$ ./dev.sh build/rasterc tests/pass/return_literal.rst --tokens
3: '@'
3: "identifier" main
3: '('
3: ')'
3: '['
4: '^'
4: "integer" 42
4: ';'
5: ']'

$ ./dev.sh build/rasterc tests/pass/return_literal.rst --ast
[1] Program
  [3] Func main
    [3] Block
      [4] Return
        [4] Int 42

$ ./dev.sh build/rasterc tests/pass/return_literal.rst --ir
; ModuleID = 'tests/pass/return_literal.rst'
source_filename = "tests/pass/return_literal.rst"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f80:128-n8:16:32:64-S128"
target triple = "x86_64-pc-linux-gnu"

define i32 @rs_main() {
entry:
  ret i32 42
}
```

5. Examples

The programs in `examples/` were run on `rasteri` and, where they are valid programs, compiled with `rasterc` and executed. The output below is what those runs printed.

5.1 Recursion and arrays: `examples/recursion_arrays.rst`

```
# Recursion and arrays: fill an array with Fibonacci numbers computed
# recursively, then compare one element with a literal array.
@fib(n) [
    ? (n < 2) [
        ^ n;
    ]
    ^ fib(n - 1) + fib(n - 2);
]

@main() [
    expected := <0, 1, 1, 2, 3, 5, 8, 13>;
    results := arr(8);
    i := 0;
    * (i < 8) [
        results[i] := fib(i);
        i := i + 1;
    ]
    print(results[7]);
    print(results[7] == expected[7]);
    ^ 0;
]
```

```
$ ./dev.sh build/rasteri examples/recursion_arrays.rst
13
1
(exit status 0)

$ ./dev.sh build/rasterc examples/recursion_arrays.rst -o build/recursion_arrays
$ ./dev.sh build/recursion_arrays
13
1
(exit status 0)
```

5.2 While loop and logic: `examples/while_logic.rst`

```
# While loop and logic: sum the numbers 1..20 that are divisible by 3 or 5,
# then show that & and | short-circuit.
@main() [
    i := 1;
    total := 0;
    * (i <= 20) [
        ? (i % 3 == 0 | i % 5 == 0) [
            total := total + i;
        ]
        i := i + 1;
    ]
    print(total);

    zero := 0;
    # The right side of & would divide by zero, but it is never evaluated.
    print(zero != 0 & 10 / zero > 1);
    # The right side of | is skipped because the left side is already true.
```

```
    print(1 | 10 / zero);
    print(!zero);
    ^ 0;
]
```

```
$ ./dev.sh build/rasteri examples/while_logic.rst
98
0
1
1
(exit status 0)

$ ./dev.sh build/rasterc examples/while_logic.rst -o build/while_logic
$ ./dev.sh build/while_logic
98
0
1
1
(exit status 0)
```

5.3 Runtime error: `examples/runtime_error.rst`

```
# Runtime error: valid output first, then an out-of-range array read.
@main() [
    values := <10, 20>;
    print(values[1]);
    ^ values[2];
]
```

Run each binary directly. `rasteri` interprets the file and stops at the error; it produces no executable. `rasterc` compiles successfully (the error can only be found while the program runs), and the executable fails the same way when started:

```
$ ./dev.sh build/rasteri examples/runtime_error.rst
20
rastera: runtime error: array index out of range
(exit status 2; the error text is on stderr)

$ ./dev.sh build/rasterc examples/runtime_error.rst -o build/runtime_error
$ ./dev.sh build/runtime_error
20
rastera: runtime error: array index out of range
(exit status 2; the error text is on stderr)
```

5.4 Static error: `examples/static_error.rst`

```
# Static error: 'missing' is never assigned, so no program is run or built.
@main() [
    print(missing);
    ^ 0;
]
```

```
$ ./dev.sh build/rasteri examples/static_error.rst
examples/static_error.rst:3: error: undefined variable 'missing'
(exit status 1)
```

```
$ ./dev.sh build/rasterc examples/static_error.rst -o build/static_error
examples/static_error.rst:3: error: undefined variable 'missing'
(exit status 1; build/static_error is not created)
```

5.5 Graphics: `examples/graphics_demo.rst`

```
# Graphics demo: a 64x64 gradient with a white rectangle and two diagonal
# lines, written to out.ppm in the current directory.
@main() [
    canvas(64, 64);

    # Red grows to the right, green grows downwards, blue is constant.
    y := 0;
    * (y < 64) [
        x := 0;
        * (x < 64) [
            pixel(x, y, rgb(x * 4, y * 4, 128));
            x := x + 1;
        ]
        y := y + 1;
    ]

    rect(8, 8, 20, 12, rgb(255, 255, 255));
    line(0, 0, 63, 63, rgb(255, 0, 0));
    line(0, 63, 63, 0, rgb(0, 255, 0));
    export();
    ^ 0;
]
```

`export()` writes `out.ppm` in the directory the program runs in. Under `./dev.sh` that is the repository root (`/work`); git ignores `*.ppm`. Both programs produce the same file:

```
$ ./dev.sh build/rasteri examples/graphics_demo.rst
$ sha256sum out.ppm
2fe2e88e8620fc46ed89a9e3e675e96dc8f7fc77526dde82f5a95bda473f9fa7  out.ppm

$ ./dev.sh build/rasterc examples/graphics_demo.rst -o build/graphics_demo
$ ./dev.sh build/graphics_demo
$ sha256sum out.ppm
2fe2e88e8620fc46ed89a9e3e675e96dc8f7fc77526dde82f5a95bda473f9fa7  out.ppm
```

To look at the picture, open `out.ppm` in any image viewer that reads PPM (for example GIMP, or the image viewer of most Linux desktops). Optionally, if ImageMagick is installed on the host, `convert out.ppm out.png` makes a PNG. The file is a 64x64 image: a gradient (red increases to the right, green downwards), a white rectangle near the top-left corner and a red and a green diagonal.

5.6 Rendering the demos: `examples/gradient.rst` and `examples/pattern.rst`

`examples/gradient.rst` is the program of section 5 of the project proposal, unchanged: a 256x256 canvas filled with range loops, a conditional statement (`? ... : ...`), `pixel` and `rgb`, then `export()`. Each pixel is white when `(x * x + y * y) % 255 > 128` and dark grey otherwise, which gives rings that interfere with each other. `examples/pattern.rst` is a 128x128 picture made with `rect`, `line` and range loops: eight nested filled squares, a fan of seventeen lines from the bottom centre to the top edge, and a white frame. Both programs write `out.ppm` (binary P6) in the directory they run in; under `./dev.sh` that is the repository root, and git ignores `*.ppm`.

Run a demo with the interpreter, or compile it and run the executable; the image is the same:

```
$ ./dev.sh build/rasteri examples/gradient.rst
$ sha256sum out.ppm
cc3c3f4999df5e7d1a18c50bb9683859bba7dbd4dc6f06cd61e56b74acdff4ef  out.ppm

$ ./dev.sh build/rasterc examples/gradient.rst -o build/gradient
$ ./dev.sh build/gradient
$ sha256sum out.ppm
cc3c3f4999df5e7d1a18c50bb9683859bba7dbd4dc6f06cd61e56b74acdff4ef  out.ppm

$ ./dev.sh build/rasteri examples/pattern.rst
$ sha256sum out.ppm
02c039e36a8310591e1499af700b59ac042c40efbd0e9fd3b57f7d4a47f0f4cb  out.ppm

$ ./dev.sh build/rasterc examples/pattern.rst -o build/pattern
$ ./dev.sh build/pattern
$ sha256sum out.ppm
02c039e36a8310591e1499af700b59ac042c40efbd0e9fd3b57f7d4a47f0f4cb  out.ppm
```

Open `out.ppm` in any image viewer that reads PPM (for example GIMP or the viewer of most Linux desktops), or, if ImageMagick is installed on the host, run `convert out.ppm out.png`. The 256x256 picture is small on a modern screen; zoom in to see the rings. `tests/pass/gfx_demo_gradient.rst` and `tests/pass/gfx_demo_pattern.rst` keep these two hashes under test in both suites, so any change that alters the drawn pixels fails `./dev.sh test`.

5.7 Sorting: `examples/insertion_sort.rst` and `examples/merge_sort.rst`

`examples/insertion_sort.rst` is the program of section 4 of the project proposal: `@insertion` is copied from it unchanged. `examples/merge_sort.rst` is the recursive merge sort the proposal describes, using `arr(n)` for its temporary buffer. Both sort the same six arrays in place through an array parameter `a[]` (arrays are passed by reference): already sorted, reverse sorted, mixed, with duplicates, one element and two elements. `@print_all` prints one array and then `-1`, which cannot be an element of these arrays, so the output shows where each array ends.

```
# Insertion sort (the proposal's section 4 program) on six arrays. `insertion` sorts the
# caller's array in place; `print_all` prints one array and then -1, which cannot be an element
# here, so the output shows where each array ends. The smallest key walks j down to -1:
# `j >= 0 & a[j] > key` short-circuits, so a[-1] is never read.
@insertion(a[], n) [
    * i :: 1..(n - 1) [
         key := a[i];
         j := i - 1;
         * (j >= 0 & a[j] > key) [
             a[j + 1] := a[j];
             j := j - 1;
         ]
         a[j + 1] := key;
    ]
    ^ 0;
]

@print_all(a[], n) [
    * i :: 0..(n - 1) [
        print(a[i]);
    ]
    print(-1);
    ^ 0;
]

@main() [
```

```
    sorted := <1, 2, 3, 4, 5>;
    insertion(sorted, 5);
    print_all(sorted, 5);

    reverse := <9, 7, 5, 3, 1>;
    insertion(reverse, 5);
    print_all(reverse, 5);

    mixed := <5, 2, 9, 1, 5, 6>;
    insertion(mixed, 6);
    print_all(mixed, 6);

    dups := <4, 2, 4, 1, 2, 4>;
    insertion(dups, 6);
    print_all(dups, 6);

    one := <7>;
    insertion(one, 1);
    print_all(one, 1);

    two := <2, 1>;
    insertion(two, 2);
    print_all(two, 2);
    ^ 0;
]
```

The merge sort keeps `@merge` (which merges `a[lo..mid]` and `a[mid+1..hi]` through `tmp` and copies the result back) and `@print_all` in the file; `@main` is the same as above with `merge_sort` in place of `insertion`. Its recursive part:

```
@msort(a[], tmp[], lo, hi) [
    ? (lo < hi) [
        mid := (lo + hi) / 2;
        msort(a, tmp, lo, mid);
        msort(a, tmp, mid + 1, hi);
        merge(a, tmp, lo, mid, hi);
    ]
    ^ 0;
]

@merge_sort(a[], n) [
    tmp := arr(n);
    msort(a, tmp, 0, n - 1);
    ^ 0;
]
```

Run each example on both binaries (output joined onto one line with `paste` to save space; both programs print the same 33 lines and exit with status 0 in both phases):

```
$ ./dev.sh build/rasteri examples/insertion_sort.rst | paste -sd' '
1 2 3 4 5 -1 1 3 5 7 9 -1 1 2 5 5 6 9 -1 1 2 2 4 4 4 -1 7 -1 1 2 -1

$ ./dev.sh build/rasterc examples/insertion_sort.rst -o build/insertion_sort
$ ./dev.sh build/insertion_sort | paste -sd' '
1 2 3 4 5 -1 1 3 5 7 9 -1 1 2 5 5 6 9 -1 1 2 2 4 4 4 -1 7 -1 1 2 -1

$ ./dev.sh build/rasteri examples/merge_sort.rst | paste -sd' '
1 2 3 4 5 -1 1 3 5 7 9 -1 1 2 5 5 6 9 -1 1 2 2 4 4 4 -1 7 -1 1 2 -1

$ ./dev.sh build/rasterc examples/merge_sort.rst -o build/merge_sort
$ ./dev.sh build/merge_sort | paste -sd' '
```

```
1 2 3 4 5 -1 1 3 5 7 9 -1 1 2 5 5 6 9 -1 1 2 2 4 4 4 -1 7 -1 1 2 -1
```

`tests/pass/sort_insertion.rst` and `tests/pass/sort_merge.rst` hold the same programs with one `expect-stdout` line for every printed value, so both suites check all 33 values.

6. Language overview

Normative rules, with every error case, are in `docs/LANGUAGE_SPEC.md`. In brief:

```
# comment to end of line
@add(a, b) [                  # function; parameters are i32
    ^ a + b;                  # return
]

@set(buf[], i, x) [           # buf[] is an array parameter, passed by reference
    buf[i] := x;
    ^ 0;
]

@main() [                     # exactly one @main(); its result is the exit status
    x := 5;                   # := creates a variable, or reassigns a visible one
    ? (x > 3) [               # if
        x := x - 1;
    ] : ? (x == 0) [          # else if
        x := 100;
    ] : [                     # else
        x := 0;
    ]
    * (x > 0) [               # while: checks first, runs while non-zero
        x := x - 1;
    ]
    * i :: 1..3 [             # range loop: i = 1, 2, 3 (inclusive, i is read-only)
        x := x + i;
    ]
    a := <1, 2, 3>;           # array literal
    b := arr(4);              # 4 zeros
    b[0] := a[2];             # element read and write
    set(b, 1, 9);             # the callee's store is visible here: b[1] is 9
    print(add(b[0], 1));      # built-in print: value and newline on stdout
    ^ 0;
]
```

Core features implemented in both `rasteri` and `rasterc`:

- Two types, never written: `i32` (wrapping 32-bit signed integer) and `array` (a reference to a fixed-length heap array of `i32`; `b := a;` aliases, it does not copy).
- Arithmetic `+ - * / %` and unary `-`; `/` and `%` truncate toward zero.
- Comparisons `== != < <= > >=` (non-associative) producing 0 or 1; `&`, `|`, `!` on integers, where `&` and `|` short-circuit.
- Variables with lexical block scope; `if`, `else if`, `else`; `while`; the inclusive range loop `* i :: a..b [ ... ]`.
- Functions with parameters, recursion, mutual recursion and calls before the definition. Functions return `i32`; arrays cannot be returned. A parameter written `buf[]` is an array passed by reference (see below).
- Arrays: `<a, b, c>`, `arr(n)`, `a[i]`, `a[i] := e;`, bounds checked at run time.
- Built-in `print(x)` and the graphics built-ins described below.

- Runtime errors (exit 2): division or modulo by zero, `-2147483648 / -1` and `% -1`, array index out of range, negative array size.
- Static errors (exit 1): syntax errors, bad characters, integer literals above 2147483647, undefined or misused variables, wrong function arity, duplicate functions or parameters, built-in names reused as names, array/integer mismatches (including an array used as an operand of +, <, unary `-` and the like, and the wrong kind of argument for a parameter), missing `@main()`.

Array parameters (`@insertion(a[], n) [ ... ]`): a parameter written with `[]` is an array passed by reference, so element stores through it are visible to the caller; nothing is copied, and the array's length travels with it, so every `a[i]` through a parameter is still bounds checked. Plain parameters stay local `i32` copies. An array parameter can be indexed, stored into, aliased (`b := a;`) and passed on to other functions. Assigning a new array to it (`a := arr(3);`) rebinds that name inside the call only; the caller's array and variable are untouched. At a call, each argument must match the kind of its parameter: passing an integer for an array parameter, or an array for an integer parameter, is the static error `function 'f' expects an array for argument 2, got an integer` (or `an integer ... got an array`). `@main` takes no parameters. The proposal's insertion sort is `tests/pass/aparam_insertion_sort.rst`.

Range loop rules (`* i :: a..b [ ... ]`): `a` and `b` are `i32` and are evaluated once, in that order, before the first iteration; the body runs for `i = a, a+1, ..., b` and not at all when `a > b`; a bound of `2147483647` terminates because `i` is compared before it is incremented. `i` belongs to the loop only: it hides an outer variable of the same name until the loop ends, and assigning to it is the static error `cannot assign to loop variable 'i'`. A `[` outside parentheses after `..` opens the body, so index an upper bound with parentheses: `* i :: 0..(v[2]) [ ... ]`.

Graphics built-ins (spec section 9; all take and return `i32`, return 0 unless noted):

Call	Effect
`canvas(w, h)`	New `w` x `h` canvas, all black; `w` and `h` must be 1..4096, else the runtime error `invalid canvas size`. Calling it again replaces the canvas.
`rgb(r, g, b)`	Returns `(r << 16) | (g << 8) | b` with each channel clamped to 0..255.
`pixel(x, y, c)`	Sets one pixel; coordinates outside the canvas are ignored; only the low 24 bits of `c` are used as `0xRRGGBB`.
`line(x0, y0, x1, y1, c)`	Bresenham line in every direction, both endpoints included, clipped to the canvas.
`rect(x, y, w, h, c)`	Filled rectangle, clipped to the canvas; draws nothing if `w` or `h` is 0 or negative.
`export()`	Writes the canvas to `out.ppm` in the current directory: binary P6, header `P6\nW H\n255\n`, then `W*H` RGB byte triples, row by row from the top-left.

`(0, 0)` is the top-left pixel. `pixel`, `line`, `rect` and `export` before any `canvas` call fail with the runtime error `no canvas` (exit 2). `rasteri` keeps its own canvas and `rasterc` programs use the one in `runtime/rastera_rt.c`; both use the same algorithms and write byte-identical files. The names of the built-ins are reserved like `print`.

7. Exit status and error summary

Situation	`rasteri FILE`	`rasterc FILE -o OUT`**, then** `./OUT`
Success	stdout output, exit status = result of `@main`	`rasterc` exits 0; `./OUT` prints and exits with the result of `@main`
Static error	`FILE:LINE: error: ...` on stderr, exit 1	same message, exit 1, no `OUT` written

Situation	`rasteri FILE`	`rasterc FILE -o OUT`, then `./OUT`
Runtime error	stdout so far, `rastera: runtime error: ...` on stderr, exit 2	`rasterc` exits 0; `./OUT` behaves as in the left column
Bad usage / missing file	usage or `FILE: No such file or directory`, exit 1	same, exit 1

8. Pipeline

```
Phase 1:  source.rst -> src/interp/lexer.l -> tokens -> src/interp/interp.y -> run @main
Phase 2:  source.rst -> src/parse/lexer.l -> tokens -> src/parse/parser.y -> AST
          -> src/sema/ -> checked AST -> src/codegen/ -> LLVM IR
          -> src/backend/ -> object file -> cc + runtime/rastera_rt.c -> executable
```

8.1 How each phase uses its Bison grammar

Both grammars use Bison's default C parser skeleton; the action code inside them is C++17, so the generated parsers are compiled as C++.

Phase 1 (`src/interp/interp.y`). Everything that runs the program is in this one Bison file: the grammar actions, the node types they create, the static checks, the evaluator and `main`. The actions do not execute statements the moment a rule is reduced. They build small execution nodes, and once the whole file has parsed, the code at the end of `interp.y` checks the program and then executes `@main` by walking those nodes. Executing directly inside the actions is not possible for this language:

- A function may be called before it is defined, so every definition has to be parsed first.
- A loop body is reduced once by the parser but must run many times; a branch that is not taken must not run at all.
- `&` and `|` must skip their right operand, but Bison reduces both operands before the operator's rule.

The nodes are a tree the program runs directly. They are not intermediate code: nothing is translated, written out or compiled.

Phase 2 (`src/parse/parser.y`). The compiler's grammar describes the same language. Its actions build an AST (`src/ast/ast.hpp`); `src/sema/` checks it and `src/codegen/` walks it to emit LLVM IR, which `src/backend/` compiles to an object file and links into an executable. IR is produced from the AST instead of inside the grammar actions for the same reasons as above, and one more: LLVM needs a function's declaration before a call to it can be emitted, and a loop's condition block must exist before the condition's instructions are placed in it. Separating parsing from code generation also lets `rasterc --ast` show the tree, and lets every semantic error be reported before any IR is written. The two phases share the language definition and the test suite, not the grammar file: both pass the same 161 test cases.

9. Regenerating README.pdf

`README.pdf` is generated from this file by a dependency-free script (Python 3 standard library only); no LaTeX or pandoc is needed. It is deterministic: the same `README.md` always produces byte-identical output.

```
python3 docs/build_readme_pdf.py           # rewrite README.pdf from README.md
python3 docs/build_readme_pdf.py --check   # exit 1 if README.pdf is out of date
```

CI runs the `--check` command on every push and pull request, so a README.md edit without a regenerated README.pdf fails the build.

10. Project file inventory

Every file tracked in the repository, by directory. Generated files (`build/`) and git-ignored files (`.gitignore` ignores `build/`, `a.out` and `*.ppm`) are not listed.

10.1 Repository root

File	Purpose
`README.md`	This document (source of `README.pdf`).
`README.pdf`	Generated PDF version of this document.
`CMakeLists.txt`	CMake build: targets `rasteri`, `rastera_rt` (runtime library) and `rasterc`; runs Flex and Bison.
`Dockerfile`	Development image: Ubuntu 24.04 with CMake, Ninja, gcc/g++, Flex, Bison, LLVM 18.
`dev.sh`	Runs a command in the container; `build` and `test` shortcuts.
`.gitattributes`	Forces LF line endings and marks `*.pdf` as binary.

10.2 `.github/`

File	Purpose
`.github/workflows/ci.yml`	Continuous integration: checks README.pdf is up to date, builds the image, builds both programs, runs both test suites.
`.github/pull_request_template.md`	Template for pull request descriptions.

10.3 `docs/`

File	Purpose
`docs/LANGUAGE_SPEC.md`	Normative language specification.
`docs/proposal.pdf`	Original project proposal.
`docs/build_readme_pdf.py`	Generator for `README.pdf` (Python 3 standard library only).

10.4 `examples/`

File	Purpose
`examples/recursion_arrays.rst`	Recursive Fibonacci stored in an array; prints `13` and `1`.
`examples/while_logic.rst`	While loop with `|`, `&`, `!` and short-circuiting; prints `98 0 1 1`.
`examples/runtime_error.rst`	Prints `20`, then fails with an array index error (exit 2).
`examples/static_error.rst`	Undefined variable; rejected before anything runs (exit 1).
`examples/graphics_demo.rst`	64x64 gradient with a rectangle and two lines; writes `out.ppm`.
`examples/gradient.rst`	The proposal's section 5 program: 256x256 rings from `(x * x + y * y) % 255`; writes `out.ppm`.
`examples/pattern.rst`	128x128 nested squares, a fan of lines and a frame, with range loops; writes `out.ppm`.
`examples/insertion_sort.rst`	The proposal's insertion sort on six arrays, printing each sorted array followed by `-1`.
`examples/merge_sort.rst`	Recursive merge sort with an `arr(n)` buffer on the same six arrays; identical output.

10.5 runtime/

File	Purpose
runtime/rastera_rt.c	C runtime linked into every compiled program: the process entry main that calls rs_main, rt_print (print), rt_alloc (array allocation with a length header and the negative-size check), rt_panic (prints the runtime error, exits 2) and the graphics built-ins rt_canvas, rt_rgb, rt_pixel, rt_line, rt_rect, rt_export over one canvas that is freed at exit.

10.6 src/ (Phase 2 compiler, rasterc)

File	Purpose
src/main.cpp	rasterc driver: argument parsing, runs the pipeline, --tokens, --ast, --ir.
src/diag.hpp	Diagnostics: collects and prints FILE:LINE: error: message.
src/parse/lexer.l	Flex lexer for the compiler.
src/parse/parser.y	Bison grammar for the compiler; builds the AST.
src/parse/frontend.hpp	Interface between driver, lexer and parser (parse, lexer_begin, token_name).
src/ast/ast.hpp	AST node types and the visitor interface.
src/ast/ast_printer.hpp	Printer behind --ast.
src/sema/sema.hpp	Semantic analyzer interface.
src/sema/sema.cpp	Scope, type and call checks over the AST.
src/sema/symbols.hpp	Symbol table and scope structures.
src/codegen/codegen.hpp	LLVM IR generator interface.
src/codegen/codegen.cpp	AST to LLVM IR, including runtime-check calls.
src/backend/backend.hpp	Object emission and linking interface.
src/backend/backend.cpp	LLVM module to object file, then link with the runtime into an executable.

10.7 src/interp/ (Phase 1 interpreter, rasteri)

File	Purpose
src/interp/lexer.l	Flex lexer for the interpreter (generated symbols prefixed ip).
src/interp/interp.y	Bison grammar, execution nodes, static checks, evaluator and main of rasteri.

10.8 tests/

File	Purpose
tests/run_tests.sh	Test runner: tests/run_tests.sh BIN (compiler) or --interp BIN (interpreter). Reads # expect-exit, # expect-stdout, # expect-error, # expect-runtime-error directives, and # expect-ppm-size, # expect-ppm-pixel, # expect-ppm-sha256 to check the out.ppm a program wrote.

tests/pass/ holds 87 programs that must run (expected stdout and exit status in directives); tests/fail/ holds 74 programs that must be rejected or fail at run time. Each runs in both suites, giving the 161 per suite.

tests/pass/ files:

File	Checks
aparam_alias_in_function.rst	b := a; inside a function aliases the caller's array (prints 8).
aparam_forward_call.rst	Array parameter function called before its definition (prints 3).

File	Checks
aparam_insertion_sort.rst	The proposal's insertion sort on <5, 2, 9, 1, 5, 6>, including j = -1 short-circuit (prints 1 2 5 5 6 9).
aparam_int_param_still_copy.rst	An i32 parameter beside an array parameter stays a copy (prints 7, 99, 5).
aparam_literal_argument.rst	<...> and arr(n) passed directly as array arguments (prints 6, 0).
aparam_modify_caller.rst	A function fills the caller's array through its a[] parameter.
aparam_pass_through.rst	An array parameter handed on to another function (prints 42, 43).
aparam_range_bounds.rst	Array parameter used in a range bound only through indexing (prints 10, 30).
aparam_rebind_local.rst	a := arr(3); rebinds the parameter only; the caller's array keeps its earlier stores.
aparam_recursive_reverse.rst	Recursion storing through an array parameter at every level (prints 6 to 1).
aparam_recursive_sum.rst	Recursive sum over an array parameter (prints 21).
aparam_two_arrays.rst	Two array parameters bound to different caller arrays; copies stay separate (prints 1 2 3 2).
arithmetic_acceptance.rst	(2 + 3) * -4 % 7 gives exit status 250.
arithmetic_left_assoc.rst	Subtraction is left-associative (exit 4).
arithmetic_precedence.rst	* binds tighter than + (exit 14).
array_alias.rst	b := a; aliases; writing b[0] changes a[0] (exit 99).
array_alloc.rst	arr(5), element stores, printing elements.
array_literal.rst	Array literal and summing its elements (exit 6).
array_store_load.rst	Store then load an element (exit 7).
array_zero.rst	arr(0) is a valid empty array.
call_in_expr.rst	print used as an expression returns 0 (prints 42).
cmp_basic.rst	All six comparison operators (exit 6).
code_after_return.rst	Unreachable statements after ^ are harmless (exit 7).
combined_array_literal_comparison.rst	Indexing an array literal inside a comparison in ? (exit 7).
combined_recursive_array_index.rst	Recursion creating an array per call (exit 6).
func_args_left_to_right.rst	Arguments evaluate left to right (prints 1, 2; exit 12).
func_call_before_definition.rst	Calling a function defined later (exit 12).
func_factorial.rst	Recursive factorial (exit 120).
func_fibonacci.rst	Recursive Fibonacci (prints 55).
func_general_return.rst	sq(x) + sq(x + 1) - 1 with calls inside expressions (exit 24).
func_mutual_recursion.rst	Mutual recursion (prints 1, 0).
func_param_is_local.rst	Assigning a parameter does not change the caller's variable.
func_params_basic.rst	Parameters passed and used (exit 5).
gfx_canvas_export.rst	canvas and export both return 0; the file has header P6 4 3 255 and is black (prints 0, 0).
gfx_canvas_max_size.rst	The largest canvas, 4096 x 4096, is accepted; a later 4096 x 1 canvas replaces it.
gfx_canvas_replace.rst	A second canvas call replaces the first: new size, black again.
gfx_demo_gradient.rst	examples/gradient.rst (the proposal's program), checked by size and sha256 of out.ppm.
gfx_demo_pattern.rst	examples/pattern.rst, checked by size, two pixels and sha256 of out.ppm.

File	Checks
gfx_line_clipped.rst	A line starting and ending outside the canvas is clipped per pixel; an entirely outside line draws nothing.
gfx_line_diagonal.rst	Both diagonals of a 5x5 canvas, one in each direction of travel.
gfx_line_horizontal.rst	Horizontal line includes both endpoints and nothing around them.
gfx_line_reversed.rst	Shallow line drawn right to left, `(6,2)` to `(0,0)`.
gfx_line_single_point.rst	Equal endpoints draw exactly one pixel.
gfx_line_steep.rst	Steep line `(0,0)` to `(2,6)` draws one pixel per row.
gfx_line_vertical.rst	Vertical line drawn bottom-up covers both endpoints.
gfx_pixel_bounds.rst	`pixel` inside and outside the canvas (no row wrap-around) and only the low 24 bits of the colour.
gfx_rect_clipped.rst	Rectangles hanging off the top-left and bottom-right are clipped.
gfx_rect_zero.rst	Zero or negative width/height draws nothing; the file is the header plus 27 zero bytes (sha256).
gfx_rgb_clamp.rst	`rgb` packs channels and clamps each to 0..255 (prints 66051, 16777215, 16711808, 0, 255).
gfx_scene_sha256.rst	A scene of rectangles, clipped lines and a pixel, checked by sha256 of the whole file.
if_branches_both_return.rst	`if`/`else` where both branches return (exit 15).
if_else.rst	`if` / `else` (exit 7).
if_elseif_chain.rst	`else if` chain; 300 becomes exit 44 (low 8 bits).
if_scope.rst	Variable declared in a branch is local; outer variable still usable (exit 10).
if_without_else.rst	`if` without `else` (exit 42).
implicit_return.rst	Reaching the end of a function returns 0.
interp_deep_recursion.rst	`fib(20)` through deep recursion (exit 109).
interp_local_array_alias.rst	Alias of a local array inside a function (prints 9).
interp_wraparound.rst	`2147483647 + 1` wraps to `-2147483648`.
logic_short_circuit.rst	`&` and `|` skip the right operand when decided (prints 3, 4; exit 3).
logic_truth_precedence.rst	Logical operator precedence and `!` (prints four 1s).
print_acceptance.rst	Basic `print` (prints 7).
print_multiple.rst	Several `print` calls in order (10, 20, 30).
range_bound_expressions.rst	Upper bounds ending in a variable, a parenthesised index and `f(x) - 1` (prints 15, 3, 9).
range_body_variable_per_iteration.rst	A variable declared in the body is new on every iteration (prints 1, 2, 3).
range_bounds_once.rst	Both bounds are calls with output; each runs once, `a` before `b` (prints 10, 20, 1, 2, 3).
range_empty.rst	`a > b` runs the body zero times, including negative bounds (prints 0, 7).
range_int_max.rst	`2147483646..2147483647` ends after two iterations (no counter overflow).
range_nested.rst	Nested range loops, the inner bound using the outer variable (prints 36, 6).
range_return_from_body.rst	`^` inside a range body returns from the function (exit 3).
range_shadows_outer.rst	The loop variable hides an outer `i`, which is unchanged afterwards (prints 5, 6, 5).
range_single.rst	`a == b` runs exactly once (prints 1, 3).
range_sum.rst	Sum 1 to 10 with a range loop (prints 55).
range_while_array.rst	Range loop fills an array, a while loop sums it (prints 55).

File	Checks
`return_literal.rst`	Returned literal becomes the exit status (42).
`rt_array_zero.rst`	`arr(0)` is created without a runtime error (prints 5).
`sort_insertion.rst`	`examples/insertion_sort.rst`: six arrays sorted in place, all 33 printed values checked.
`sort_merge.rst`	`examples/merge_sort.rst`: recursive merge sort with an `arr(n)` buffer, same 33 values.
`unary_minus.rst`	`-(-7)` is 7 (exit 7).
`var_basic.rst`	Variable creation and use (exit 5).
`var_inner_block_reassigns_outer.rst`	A bare block reassigns a visible outer variable (exit 2).
`var_inner_declared_fresh_after_block.rst`	A name first created inside a block is new again afterwards (exit 4).
`var_reassign_arith.rst`	Reassigning with arithmetic (exit 7).
`while_nested_multiplication.rst`	Nested while loops (prints 6).
`while_return_from_body.rst`	`^` inside a loop body returns from the function (exit 23).
`while_sum_1_to_10.rst`	Sum 1 to 10 with `while` (prints 55).
`while_zero_iterations.rst`	A false condition runs the body zero times (prints 0).

`tests/fail/` files (expected stderr text in parentheses; static errors exit 1, runtime errors exit 2):

File	Checks
`aparam_arithmetic_on_array.rst`	Array parameter used in `a + 1` (`operands must be integers`).
`aparam_array_for_int.rst`	Array passed where an integer parameter is expected (`function 'first' expects an integer for argument 1, got an array`).
`aparam_assign_int_to_array.rst`	Integer assigned to an array parameter (`type mismatch: cannot assign integer to variable 'a' of type array`).
`aparam_bounds_read.rst`	Read past the end through a parameter (runtime: `array index out of range`, output 1 kept first).
`aparam_bounds_store.rst`	Store through a nested array-parameter call past the end (runtime: `array index out of range`, output 7 kept first).
`aparam_builtin_name.rst`	Array parameter named `print` (`using built-in name 'print' as a parameter name`).
`aparam_duplicate.rst`	`@f(a, a[])` (`duplicate parameter 'a'`).
`aparam_index_int_param.rst`	Indexing an integer parameter (`cannot index a non-array value`).
`aparam_int_for_array.rst`	Integer passed where an array parameter is expected (`function 'first' expects an array for argument 1, got an integer`).
`aparam_int_param_as_array_store.rst`	Element store through an integer parameter (`cannot index a non-array value`).
`aparam_main_array_param.rst`	`@main(a[])` (`@main() must not have parameters`).
`aparam_negate_array.rst`	Unary `-` on an array parameter (`operand must be an integer`).
`aparam_return_array.rst`	Returning an array parameter (`return value must be an integer`).
`aparam_second_arg_kind.rst`	Integer for the second, array, parameter (`function 'f' expects an array for argument 2, got an integer`).
`aparam_wrong_arity.rst`	Too few arguments for `@f(a[], n)` (`wrong argument count for 'f': expected 2, got 1`).
`array_arith.rst`	Array nested inside an array literal `<1, <2, 3>>` (`array literal elements must be integers`).
`array_compare.rst`	Comparing two arrays with `==` (`operands must be integers`).
`array_index_non_array.rst`	Indexing an integer (`cannot index a non-array value`).

File	Checks
array_negative_size.rst	arr with a negative size (runtime: negative array size).
array_print_array.rst	print of an array (expects an integer argument).
array_type_mismatch.rst	Reassigning an integer variable with an array (type mismatch).
bad_character.rst	Unknown character $ (unexpected character '$').
call_arg_undefined.rst	print(y) with y never assigned (undefined variable 'y').
call_unknown_func.rst	Call to an undefined function (unknown function 'unknown_func').
call_wrong_arity_too_few.rst	print() (wrong argument count for 'print').
call_wrong_arity_too_many.rst	print(1, 2) (wrong argument count for 'print').
cmp_nonassoc.rst	Chained comparison a < b < c (syntax error).
func_builtin_name.rst	Function named print (using built-in name 'print' as a function name).
func_duplicate.rst	Two functions with one name (duplicate function 'main').
func_duplicate_param.rst	Repeated parameter name (duplicate parameter 'a').
func_main_with_params.rst	@main with parameters (@main() must not have parameters).
func_param_builtin_name.rst	Parameter named print (... as a parameter name).
func_vars_not_shared.rst	A function using another function's variable (undefined variable 'x').
func_wrong_arity.rst	Wrong number of arguments (expected 2, got 1).
gfx_array_argument.rst	Array passed to rgb (builtin 'rgb' expects an integer argument).
gfx_canvas_invalid_stops.rst	canvas(4, 5000) stops the program after output 5 (runtime: invalid canvas size).
gfx_canvas_negative.rst	Negative canvas height (runtime: invalid canvas size).
gfx_canvas_too_large.rst	Canvas width 4097 (runtime: invalid canvas size).
gfx_canvas_zero.rst	Canvas width 0 (runtime: invalid canvas size).
gfx_export_arity.rst	export(1) (wrong argument count for 'export').
gfx_export_no_canvas.rst	export() before any canvas, after printing 1 (runtime: no canvas).
gfx_function_builtin_name.rst	Function named canvas (using built-in name 'canvas' as a function name).
gfx_line_arity.rst	line with four arguments (wrong argument count for 'line').
gfx_line_no_canvas.rst	line before any canvas (runtime: no canvas).
gfx_param_builtin_name.rst	Parameter named pixel (using built-in name 'pixel' as a parameter name).
gfx_pixel_no_canvas.rst	pixel before any canvas (runtime: no canvas).
gfx_rect_no_canvas.rst	rect before any canvas (runtime: no canvas).
gfx_rgb_variable_name.rst	Variable named rgb (using built-in name 'rgb' as a variable name).
if_branch_var_not_visible.rst	Variable created inside an if branch used after it (undefined variable 'branch_var').
interp_output_before_runtime_error.rst	Output 7 is kept, then runtime error negative array size.
literal_too_large.rst	Literal 2147483648 (integer literal out of range).
logic_array_not_operand.rst	! applied to an array (logical operand must be an integer).
logic_array_operand.rst	&/| applied to an array (logical operands must be integers).
no_main.rst	No @main (program has no @main() function).
range_array_bound.rst	Array as a range bound (range bound must be an integer).

File	Checks
range_assign_loop_variable.rst	Assignment to the loop variable (cannot assign to loop variable 'i').
range_index_bound_needs_parens.rst	0..v[2] [reads [as the body (syntax error).
range_variable_not_visible.rst	Loop variable used after the loop (undefined variable 'i').
rt_division_by_zero.rst	Division by zero after printing 7 (division by zero).
rt_division_int_min.rst	-2147483648 / -1 (integer division overflow).
rt_index_read_high.rst	Read past the end (array index out of range).
rt_index_read_negative.rst	Read at a negative index (array index out of range).
rt_index_store_high.rst	Store past the end (array index out of range).
rt_modulo_by_zero.rst	Modulo by zero (modulo by zero).
rt_modulo_int_min.rst	-2147483648 % -1 (integer modulo overflow).
rt_negative_array_size.rst	Negative arr size at run time (negative array size).
syntax_missing_semicolon.rst	Missing ; (3: error: syntax error, unexpected ']').
syntax_unclosed_paren.rst	Unclosed parenthesis (syntax error).
var_builtin_name.rst	Variable named print (using built-in name 'print' as a variable name).
var_inner_not_visible.rst	Block variable used after the block (4: error: undefined variable 'y').
var_self_reference_undeclared.rst	x := x + 1; with no earlier x (undefined variable 'x').
var_undefined.rst	Use of an unassigned variable (undefined variable 'z').
while_array_condition.rst	Array as a while condition (while condition must be an integer).
while_body_variable_not_visible.rst	Loop-body variable used after the loop (undefined variable 'hidden').